Analysis of Pagerank Algorithm and HITS algorithm using citation network data

Sagnik Chakravarty
VIT Chennai
sagnik.chakravarty@vitstudent.ac.in

Swapneel Chanda
VIT, Chennai
swapneel.chanda@vitstudent.ac.in

Dr Saroj Kumar Dash
Asst. Professor
Dept. Mathematics
VIT Chennai, Tamil Nadu 600127
sarojkumar.dash@vit.ac.in

We all like to rank things in our life be it movies, games, marks so on and so forth, so why not internet? During internets infancy when a term was searched all the results would pop up without any order or relevance to what the user desired, thus a need for ranking arose in this sort of scenarios, thus comes page rank algorithm developed by Sergey Brin and Larry Page and HITS algorithm which was developed by Jon Kleinberg in the late 90's. In this paper we are going to compare HITS and pagerank algorithm on Citation network dataset with around 23.3k nodes. We will be checking on the efficiency of both the ranking algorithms and then we will be checking on the spearman rank correlation for both the models tom see how closely correlated are they. The paper will also focus on the network analysis of the citation network dataset.

1 keywords

Pagerank algorithm, HITS algorithm, Citation Network Analysis, Analysis between Pagerank and HITS

Nomenclature

1. PR(A): is the Page Rank of page A
2. PR(Ti): is the Page Rank of pages Ti which link to page A
3. C(Ti): is the number of outbound links on page Ti and
4. d: is a damping factor which can be set between 0 and 1 optimal is 0.85
5. M: adjacency matrix
6. M^i: adjacency matrix after i^{th} iteration
7. u: hub weight vector
8. v: authority weight vector

2 Introduction

As the complexity of internet has increased over time the reason to rank each of the individual page in order of some criteria has also increased as it would be hard to navigate the web without any form of guide as a user can be bombarded by many unhelpful links. That is why the ranking of web page was developed and two of the most popular method of those ranking came to be as page rank algorithm and hits algorithm.

In our paper we have tackled this issue head on first we have tried to create a dummy network with 20 nodes and ran pagerank, HITS algorithm and random walk on it and w=found the spearman rank correlation between the ranks to check whether there is any correlation between the ranking or not. We have deployed random walk in this case with having the idea that if we simulate a point jumping from one node to another on basis of how one node directs it to another node and ran the simulation for a large number of times then we can get the probability for each of the nodes and thus we can get an estimated ranking for each of the nodes. We will then use this ranking as a form of control to check against the ranking we get from page rank and HITS algorithm to compare each of their correlation and to see which among the two is closer to the ones we obtain from random walk to check which is better in the long run.

We will then apply this model on a citation network graph with 28.1k nodes[1] and see how our assumption performs against real life dataset.

3 Literature Review

1. **Pagerank algorithm:** This algorithm was developed in 1998 by Sergey Brin and larry page [2]. Pagerank algorithm employs a technique by which the importance of a webpage is measured by the number of inlinks nd outlinks from that webpage. The numerical weight that

it assigns to any given element E is referred to as the PageRank of E and denoted by PR(E).
The very basic Model for Page rank algorithm is used as the Random Surfer Model where a the user clicks on content with no regard towards the content.

$$PR(A) = (1-d) + d * (\frac{PR(T_i)}{c(T_i)} + \cdot + \frac{PR(T_n)}{c(T_n)}) \quad (1)$$

Example: Consider the network below

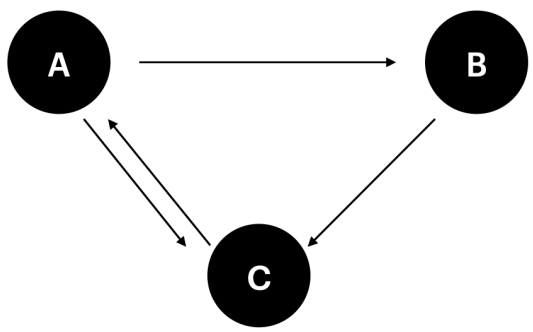

Fig. 1. Network graph with 3 nodes

The Initial Page ranks are always assumed to be 1 for all web page
PR(A)=(1-d)+d(PR(Ti/C(Ti)+. . . .+PR(Tn)/C(Tn))
PR(A)=1-d+d(PR(C)/C(C))=0.15+0.85[1/1]=1
PR(B) = 1-d +d[PR(A)/C(A)] = 0.15 + 0.85[1/2] = 0.575
PR(C) = 1-d + d[PR(A)/C(A) + PR(B)/C(B)] = 0.15 + 0.85[1/2 + 1/1] = 1.425

Matrix Representation of Pagerank algorithm Consider the previous graph the graph for the previous have an adjacency matrix as:

$$\begin{array}{cccc} & A & B & C \\ \text{M} = A & 0 & 1/2 & 1/2 \\ B & 0 & 0 & 1 \\ C & 1 & 0 & 0 \end{array}$$

$$\text{M}^1 = 1 - d + M^T * d = \begin{bmatrix} 0.15 \\ 0.15 \\ 0.15 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{bmatrix} * \begin{bmatrix} 0.85 \\ 0.85 \\ 0.85 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.575 \\ 1.425 \end{bmatrix}$$

M^2 = 1-d + M^T*M^1
we will continue till convergence
code for Pagerank algorithm

```
import numpy as np

# Define the adjacency matrix
adj_mat = np.array([[0, 1, 1], [1, 0,
    1], [1, 1, 0]])

# Define the damping factor (usually set
    to 0.85)
damp_factor = 0.85

# Compute the PageRank scores using the
    power method
n = adj_mat.shape[0]
v = np.ones(n) / n # initial value for
    page rank
last_v = np.ones(n) * 10**(-9) # for
    convergence
M = adj_mat / adj_mat.sum(axis=0)
while np.linalg.norm(v - last_v, 2) >
    10**(-9):
   last_v = v.copy()
   v = damp_factor * M @ v + (1 -
       damp_factor) / n

# Print the PageRank scores
for i, score in enumerate(v):
   print(f"Node {i} has a PageRank score
       of {score}")
```

Output

```
Node 0 has a PageRank score of
    0.3333333333333333
Node 1 has a PageRank score of
    0.3333333333333333
Node 2 has a PageRank score of
    0.3333333333333333
```

Strength of Pagerank
1) PageRank is advantageous over the HITS method because it has a lower query-time cost when incorporating a page's precomputed PageRank significance score.

2) PageRank is less susceptible to the spam of specialized links because it is calculated using the entire Web graph, not just a small subset.

3) PageRank is more efficient than HITS algorithm because it computes a single metric score during crawl time and blends it with informational retrieval system during query time.

4) The PageRank algorithm is more practical in the current environment because it computes at crawl time, unlike HITS, which requires computation at query time. However, the choice of algorithm depends on specific use cases and the trade-offs between efficiency and accuracy.

Drawback of pagerank
1) Rank Sinks occur in a network when pages get stuck in an infinite loop of links.

2) Spider Traps are another issue that can arise with PageRank. They occur when a group of pages has no

links to pages outside the group, creating a dead end.

3) A link on a website that points to a page with no outbound links is known as a Dangling Link, which can decrease a page's PageRank.

4) Pages with no outgoing links are referred to as Dead Ends, which can negatively impact a website's PageRank.

5) PageRank ignores pages with no outbound links because they can decrease the PageRank of other pages on the site.

6) Circular References, or links that create an infinite loop, can also lower the PageRank of a website.

7) Adding a page to a website can raise its PageRank by 0.428, but it can also lower the PageRank of other pages on the site. Swapping links with high PageRank websites or creating a page with high PageRank and linking it to the top page can help avoid this issue.

8) The PageRank score of a page is not influenced by its relevance to a specific query.[5]

2. **HITS algorithm:** HITS, or Hyperlink-Induced Topic Search, is.a search algorithm by which we measure two scores "authority score" and "hub score', where authority score is how reliable or authoritative the links coming out from that website are and hub score is the score of how many good or reliable in-links which are connected to that website.

   HITS is a link analysis algorithm developed by Jon Kleinberg that rates web pages. A good hub page is one that points to many other pages, while a good authority page is linked by many different hub pages.

   To determine the best hub and authority for a given adjacency matrix, the HITS algorithm can be used to calculate the hub and authority scores for k = 3. The adjacency matrix is used in this process.

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

with nodes $M = \begin{array}{c|cccc} & n1 & n2 & n3 & n4 \\ \hline n1 & 0 & 1 & 1 & 1 \\ n2 & 0 & 0 & 1 & 1 \\ n3 & 1 & 0 & 0 & 1 \\ n4 & 0 & 0 & 0 & 1 \end{array}$ Rank using out degree and in degree

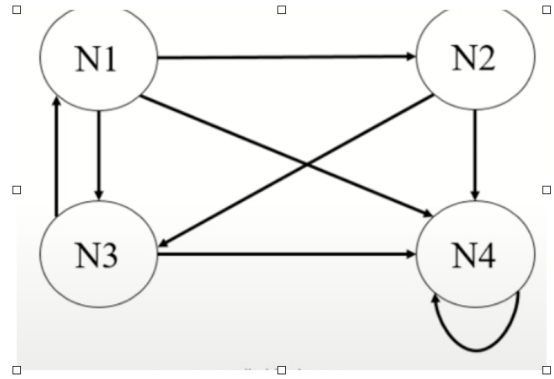

Fig. 2. Network graph with 4 nodes

Nodes	Out Degree(Hub)	In Degree(authority)
n1	3	1
n2	2	1
n3	2	2
n4	1	4

HUB: N1, N2, N3TIE, N4 AUTHORITY: N4, N3, N2, N1TIE

Now to find the rank using HITS algorithm we would be using the formula v = M^T*u where M is the adjacency matrix and u is the hub weight vector in this case we assume it to be 1, and v is the calculated authority weight vector. $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix}$

Now the new hub vector is going to be calculated as u = M^T*v $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \\ 5 \\ 4 \end{bmatrix}$

Now this are all for K = 1 to calculate to K = 2 and further we need to divide the vector u and v by their respective norm till we reach congruency Now for K = 2 u = $\frac{7}{\sqrt{126}}, \frac{6}{\sqrt{126}}, \frac{5}{\sqrt{126}}, \frac{4}{\sqrt{126}}$

which equates to u = 0.623, 0.535,0.445,0.356

We will now find authority scores

v = $\frac{1}{\sqrt{22}}, \frac{1}{\sqrt{22}}, \frac{2}{\sqrt{22}}, \frac{4}{\sqrt{22}}$ We will observe that the values will converge after K = 2 hence the ranking we get is

Nodes	Out Degree(Hub)	In Degree(authority)
N1	0.613	0.213
N2	0.535	0.213
N3	0.445	0.426
N4	0.356	0.853

HUB: N1,N2,N3,N4 AUTHORITY: N4,N3,N2,N1(TIE)

code for HITS

```
import numpy as np

# Define the adjacency matrix
```

```
adj_mat = np.array([[0, 1, 1], [1, 0,
    1], [1, 1, 0]])

# Define the number of iterations
num_iters = 20

# Compute the HITS scores using the
    power method
n = adj_mat.shape[0]
a = np.ones(n) / n # initial value for
    authority score
h = np.ones(n) / n # initial value for
    hub score
for _ in range(num_iters):
   h = adj_mat @ a
   h = h / np.linalg.norm(h, 2)
   a = adj_mat.T @ h
   a = a / np.linalg.norm(a, 2)

# Print the HITS scores
for i, (hub_score, authority_score) in
    enumerate(zip(h, a)):
   print(f"Node {i} has a hub score of
       {hub_score} and an authority
       score of {authority_score}")
```

Output

```
 Node 0 has a hub score of
     0.5773502691896258 and an authority
     score of 0.5773502691896258
Node 1 has a hub score of
    0.5773502691896258 and an authority
    score of 0.5773502691896258
Node 2 has a hub score of
    0.5773502691896258 and an authority
    score of 0.5773502691896258
```

advantage of HITS algorithm

1) One advantage of HITS is its ability to rank pages based on their relevance to the search term, resulting in pertinent hub and authority pages.
2) HITS scores can be combined with other rankings that are based on information retrieval.
3) Compared to PageRank, HITS provides a more targeted response to user queries.
4) HITS identifies important pages using the values of hubs and computed authorities.
5) HITS constructs a Web graph by identifying a set of pages that match a specific query string.

disadvantage of HITS algorithm

1) The greatest disadvantage of HITS algorithms its, the dependency on query time which makes the algorithm slow.

2) Irrelevant authorities: Sometimes, due to errors made by website creators, the scores or ratings of authorities and hubs may increase. HITS assumes that a user intentionally links a hyperlink from their webpage to another authority webpage because they genuinely believe that the authority webpage is linked to their page in some way (hub).

3) Irrelevant hubs: A page with links to many different topics can receive a high hub score while being unrelated to the given query. Despite not being the most pertinent source, this page can still have a very high hub rank for any information if it links to highly regarded authorities.

4) Mutual reinforcement of host relationships: HITS emphasizes the reciprocal support of hub webpages and authorities. A good authority is a page that is pointed to by numerous good hubs, while a good hub is a page that is pointed to by numerous good authorities.[6]

3. **Random Walk:** A path made up of a series of randomly chosen steps is referred to mathematically as a random walk. The phrase is frequently used to refer to a variable's movement over time, such as the price of a stock or the location of a particle, where each step's change in the variable is decided by chance. For instance, if you were to randomly walk across a grid of squares, you would randomly decide which way to go one step at a time (up, down, left or right). Your final course of action is a haphazard stroll. The random walk is often utilized in numerous disciplines, including physics, economics, computer science, and more.[7]

```
import numpy as np

# Adjacency matrix
adj_mat =
    np.array([[0,1,1],[1,0,1],[1,1,0]])

# Number of iterations
num_iterations = 100

# Start node
start_node = 0

# Initialize an array to store the
    probability of visiting each node
probabilities =
    np.zeros(adj_mat.shape[0])

# Set the probability of starting at the
    start node to 1
probabilities[start_node] = 1

# Perform the random walk
for i in range(num_iterations):
   # Update the probabilities by
       multiplying by the adjacency
       matrix
   probabilities = np.dot(probabilities,
       adj_mat)
   # Normalize the probabilities
   probabilities /= np.sum(probabilities)
```

```
print(probabilities)
```

Output

```
[0.33333333 0.33333333 0.33333333]
```

inference: From the random walk, Hits and Page rank algorithm we can observe that rank of each of the webpage are same to further solidify our result we can use spearman rank correlation test between page rank also and random walk and hits and random walk to compare the correlation.

4 Methodology

We will first implement the above described technique on a 20 nodes generated network then we will use the same algorithm on a the citation network with 28.1k nodes.

4.1 Implementation on a 20 nodes generated network

Code to generate 20 node network

```
library(igraph)
adj_mat = matrix(sample(0:1, 20,
    replace=TRUE), byrow = T, nrow = 20)
print(adj_mat)
g<- graph_from_adjacency_matrix(adj_mat)
summary(g)
plot(g, edge.arrow.size = .5)
```

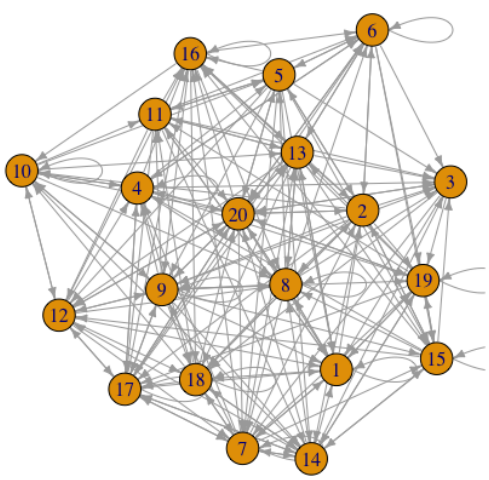

Fig. 3. Network graph with 20 nodes

Getting the ranks with both pagerank algorithm and Hits algorithm alongside random walk

```
import random
import scipy.stats as ss
import pandas as pd

# Define the adjacency matrix
lst = list(random.randint(0,1) for i in
    range(400))
adj_mat = np.array(lst).reshape(20, 20)
matlst = adj_mat.flatten()
s = str(matlst)
s = s.replace(' ', ',')
print(s)

# Define the damping factor (usually set to
    0.85)
damp_factor = 0.85

# Compute the PageRank scores using the
    power method
n = adj_mat.shape[0]
v = np.ones(n) / n # initial value for page
    rank
last_v = np.ones(n) * 10**(-9) # for
    convergence
M = adj_mat / adj_mat.sum(axis=0)
while np.linalg.norm(v - last_v, 2) >
    10**(-9):
   last_v = v.copy()
   v = damp_factor * M @ v + (1 -
       damp_factor) / n

ranks = ss.rankdata(v)
print(adj_mat)

# Print the PageRank scores
for i, score in enumerate(v):
   print(f"Node {i} has a PageRank score of
       {score}")
for i, ranks in enumerate(ranks):
# Define the number of iterations
num_iters = 1000

# Compute the HITS scores using the power
    method
n = adj_mat.shape[0]
a = np.ones(n) / n # initial value for
    authority score
h = np.ones(n) / n # initial value for hub
    score
for _ in range(num_iters):
   h = adj_mat @ a
   h = h / np.linalg.norm(h, 2)
   a = adj_mat.T @ h
   a = a / np.linalg.norm(a, 2)
rankh = ss.rankdata(h)
ranka = ss.rankdata(a)
# Print the HITS scores
for i, (hub_score, authority_score) in
    enumerate(zip(h, a)):
   print(f"Node {i} has a hub score of
```

```
        {hub_score} and an authority score
        of {authority_score}")

for i, (hub_rank, authority_rank) in
    enumerate(zip(rankh, ranka)):
   print(f"Node {i} has a hub rank of
        {hub_rank} and an authority score of
        {authority_rank}")
# Number of iterations
num_iterations = 10000
# Start node
start_node = 0
# Initialize an array to store the
    probability of visiting each node
probabilities = np.zeros(adj_mat.shape[0])
# Set the probability of starting at the
    start node to 1
probabilities[start_node] = 1
# Perform the random walk
for i in range(num_iterations):
   # Update the probabilities by
        multiplying by the adjacency matrix
   probabilities = np.dot(probabilities,
        adj_mat)
   # Normalize the probabilities
   probabilities /= np.sum(probabilities)
rankrand = ss.rankdata(probabilities)
print(probabilities)
print("random walk rank")
for i, rank in enumerate(rankrand):
   print(f"Node {i} is ranked at {rank}")


rank_authority = pd.series(ranka)
rank_hub = pd.series(rankh)
rank_RandomWalk = pd.series(rankrand)
rank_pagerank = pd.Series(rank)
print("Correlation between page rank and
    rank of hub",
    ss.pearsonr(rank_pagerank, rank_hub)[0])
print("Correlation between page rank and
    rank of authority",
    ss.pearsonr(rank_pagerank,
    rank_authority)[0])
print("Correlation between rank of
    authority and rank of hub",
    ss.pearsonr(rank_authority,
    rank_hub)[0])
print("Correlation between Random Walk and
    rank of hub",
    ss.pearsonr(rank_RandomWalk,
    rank_hub)[0])
print("Correlation between Random Walk and
    rank of authority",
    ss.pearsonr(rank_RandomWalk,
    rank_authority)[0])
print("Correlation between random walk and
    page rank",
    ss.pearsonr(rank_RandomWalk,
    rank_pagerank)[0])
```

Output

```
Correlation between page rank and rank of
    hub 0.8481203007518797
Correlation between page rank and rank of
    authority -0.3308270676691729
Correlation between rank of authority and
    rank of hub -0.2571428571428572
Correlation between Random Walk and rank of
    hub -0.1624060150375941
Correlation between Random Walk and rank of
    authority 0.8736842105263156
Correlation between random walk and page
    rank -0.24210526315789474
```

Inference As we can see that the correlation between hub rank and pagerank is highly positive that is they are highly correlated while that of random walk and page rank is close to zero hence we can say that their is no linear correlation between them definitely same case for hub rank and random walk while random walk and authority rank are highly correlated hence we can say in the case for 20 nodes HITS is a better algorithm.

4.2 implementation of the above method for citation network dataset

In this case we are implementing the above method on a citation network dataset we obtained from Network Repository [1].

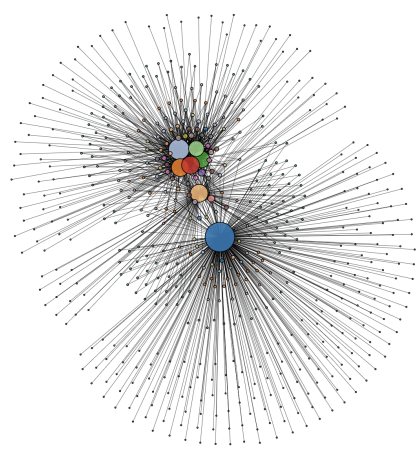

Fig. 4. Citation Network Node diagram

4.2.1 Ranking on citation network

From running the graph on the above implementation we get the spearman out put as:

```
Correlation between page rank and rank of
    hub 0.968
Correlation between page rank and rank of
    authority 0.761
Correlation between rank of authority and
    rank of hub -0.452
```

```
Correlation between Random Walk and rank of
    hub -0.008
Correlation between Random Walk and rank of
    authority 0.891
Correlation between random walk and page
    rank -0.242
```

5 Pagerank EDA

The below table shows the top 6 citation paper based citation network data.

edge	pagerank vector	rank
24689	0.0010026254	1
26858	0.0007643516	2
25232	0.0007337627	3
24073	0.0006779409	4
25254	0.0006458079	5
25237	0.0006165334	6

From the table it can be seen that 24689 is the most important citation according to page rank algorithm that is the link has the most number of inlinks and outlinks emerging from that node

basics statistics data for pagerank vector

	edge	pagerank vector
nobs	2.809300e+04	28093.000000
NAs	0.000000e+00	0.000000
Minimum	1.000000e+00	0.000010
Maximum	2.809300e+04	0.001003
1. Quartile	7.024000e+03	0.000016
3. Quartile	2.107000e+04	0.000040
Mean	1.404700e+04	0.000036
Median	1.404700e+04	0.000023
Sum	3.946224e+08	1.000000
SE Mean	4.838560e+01	0.000000
LCL Mean	1.395216e+04	0.000035
UCL Mean	1.414184e+04	0.000036
Variance	6.577040e+07	0.000000
Stdev	8.109895e+03	0.000038
Skewness	0.000000e+00	5.546117
Kurtosis	-1.200128e+00	60.148579

From the scatter plot we can see that most of the vector values are near 0 which can be obvious from the diagram as most of them do not have many inlinks but some have too many inlinks as can be seen from the scattered diagram.

page rank boxplot also support our assertion that the data is tightly packed

We will next fit a distribution for the curve.

from the graph the pagerank values follow the gamma distribution with parameter 1.91 and 53754867

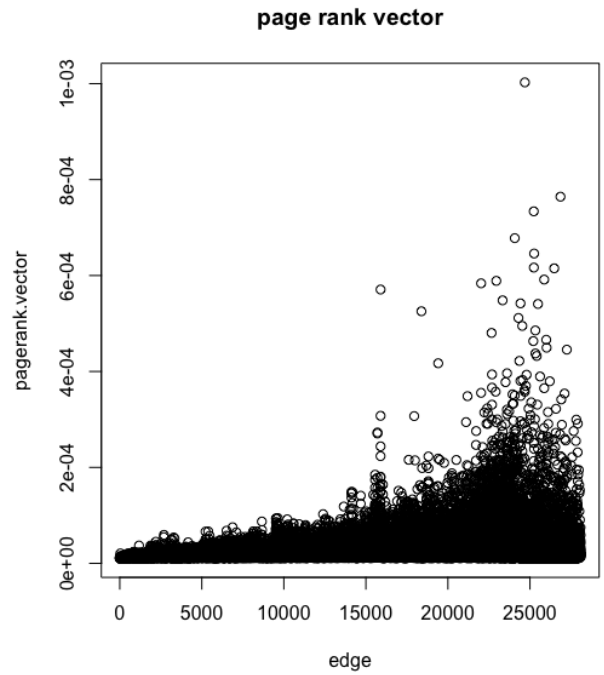

Fig. 5. Pagerank Vector Scatter Plot

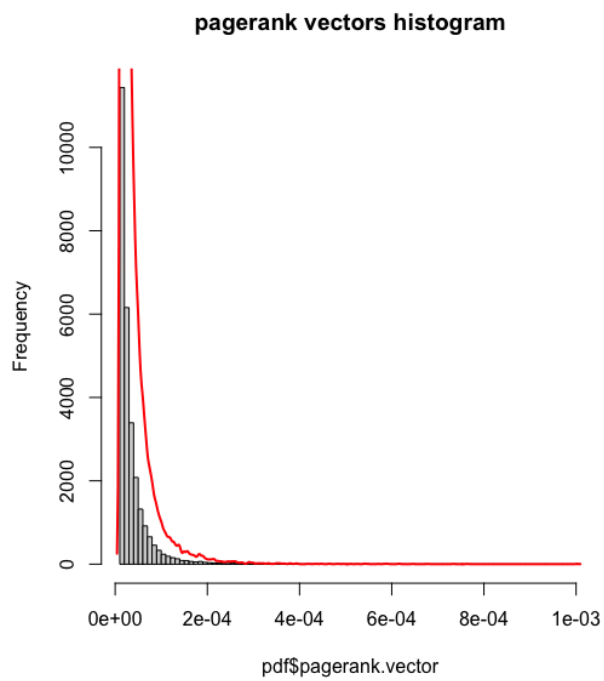

Fig. 6. Pagerank Historgram

6 Authority Rank EDA

The below table shows the top 6 citation paper based on citation network data.

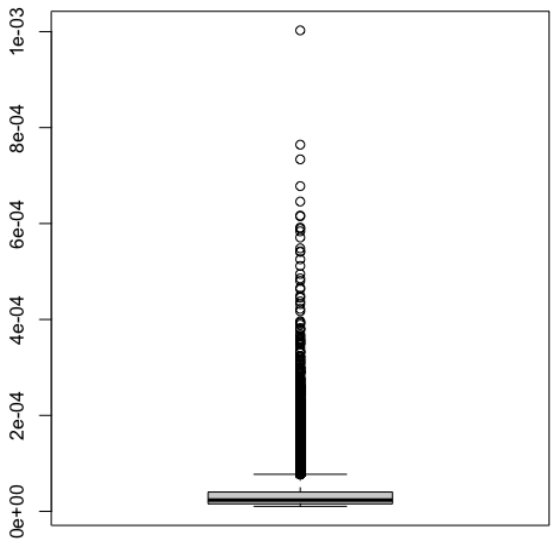

Fig. 7. Pagerank BoxPlot

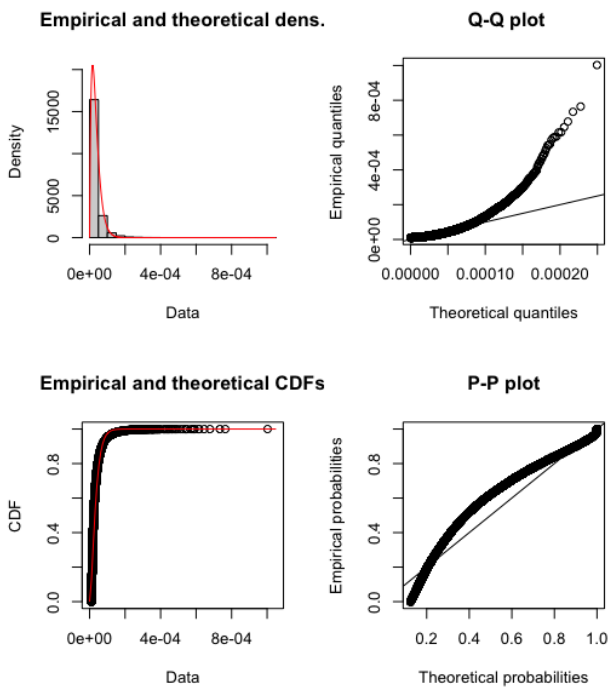

Fig. 8. Pagerank Distribution Fit

edge	authority rank	rank
2692	1.0000000	1
2686	0.7864863	2
3077	0.7298729	3
2663	0.7259115	4
3992	0.7145666	5
157	0.6744820	6

From the table it can be seen that 2692 is the most important citation according to HITS algorithm that is the link has the most number of reliable outlinks emerging from that node

basics statistics data for Authority vector

	edge	authority rank
nobs	2.809300e+04	28093.000000
NAs	0.000000e+00	0.000000
Minimum	1.000000e+00	0.000000
Maximum	2.809300e+04	1.000000
1. Quartile	7.024000e+03	0.000235
3. Quartile	2.107000e+04	0.009344
Mean	1.404700e+04	0.013132
Median	1.404700e+04	0.001883
Sum	3.946224e+08	368.927466
SE Mean	4.838560e+01	0.000230
LCL Mean	1.395216e+04	0.012682
UCL Mean	1.414184e+04	0.013583
Variance	6.577040e+07	0.001482
Stdev	8.109895e+03	0.038502
Skewness	0.000000e+00	7.690865
Kurtosis	-1.200128e+00	89.369029

The variance is low but its still higher than pagerank which suggest that the authority vector are more scatteredly distributed that is there are more outlinks emerging than the inlinks and outlinks ration

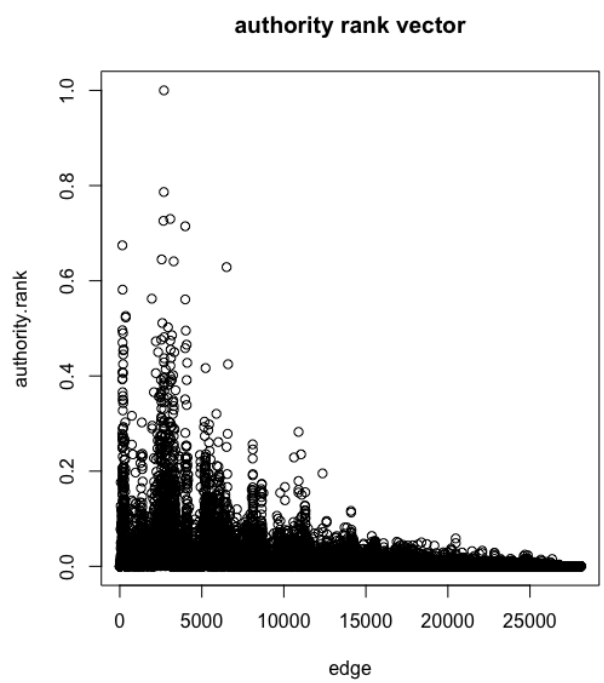

Fig. 9. Authority Rank Scatter Plot

From the scatter plot we can see that most of the vector values are near 0 which can be obvious from the diagram as most of them do not have many inlinks but some have too many inlinks as can be seen from the scattered diagram. We

then plot the boxplot for the data after removing the outliers

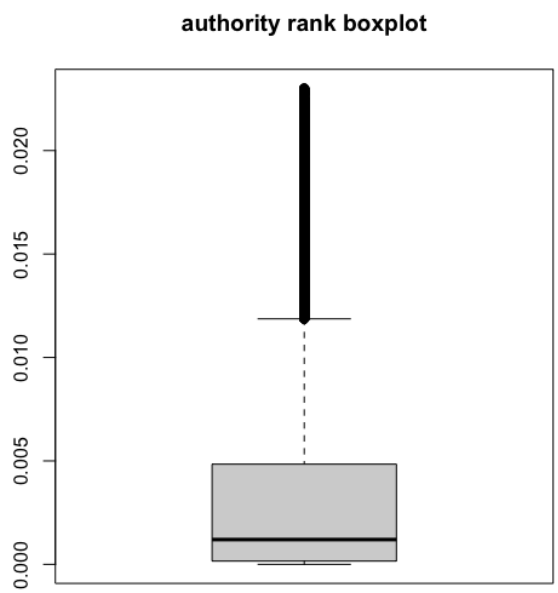

Fig. 10. Authority rank Boxplot

From the data we can see that most of the data is scattered above the median value of the authority rank which proves that the data is left skewed in nature which can also be shown from the histogram as shown below.

We will next fit a distribution for the curve.

The vector have gamma distribution with parameter shape- 0.4982727 rate- 139.1576475

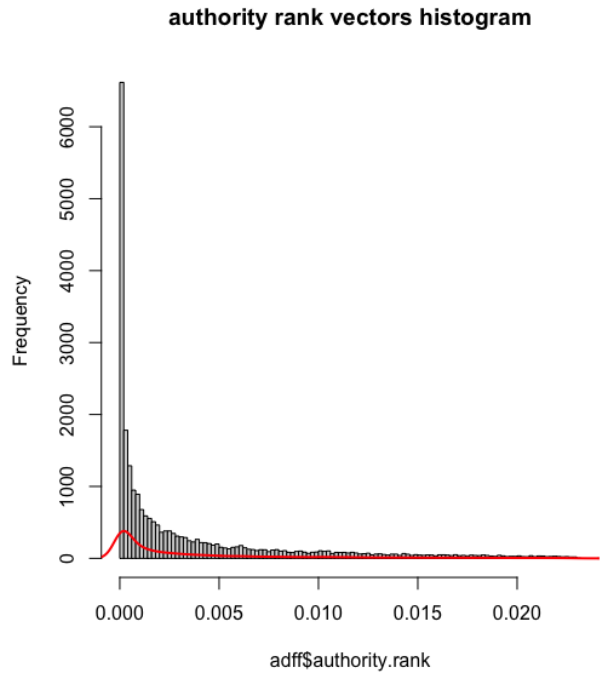

Fig. 11. Authroity Rank histogram

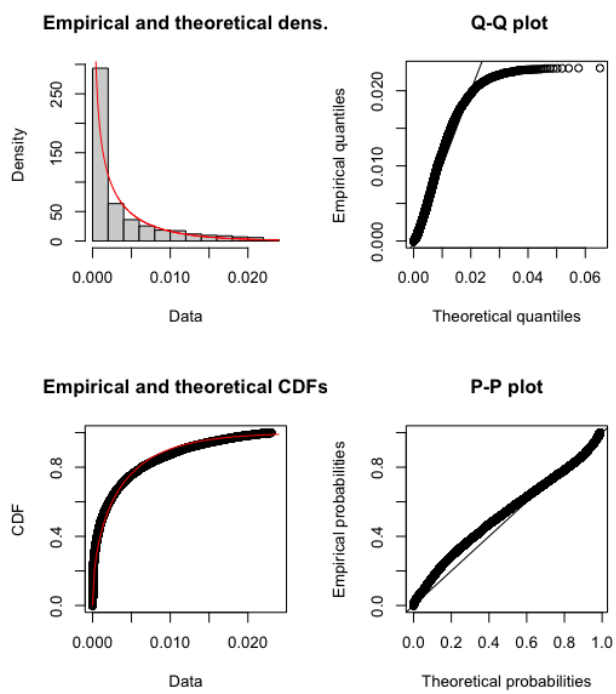

Fig. 12. Authority rank Distribution Fit

7 Hub rank EDA

The below table shows the top 6 citation paper based citation network data.

edge	hub rank	rank
152	1.0000000	1
143	0.9525030	2
157	0.8269075	3
1339	0.7793302	4
2686	0.7493091	5
48	0.7314524	6

From the table it can be seen that 152 is the most important citation according to page rank algorithm that is the link has the most number of inlinks incoming to that node

basics statistics data for pagerank vector

	edge	hub rank vector
nobs	2.809300e+04	28093.000000
NAs	0.000000e+00	0.000000
Minimum	1.000000e+00	0.000000
Maximum	2.809300e+04	1.000000
1. Quartile	7.024000e+03	0.000001
3. Quartile	2.107000e+04	0.000753
Mean	1.404700e+04	0.005303
Median	1.404700e+04	0.000038
Sum	3.946224e+08	148.982702
SE Mean	4.838560e+01	0.000171
LCL Mean	1.395216e+04	0.004968
UCL Mean	1.414184e+04	0.005639
Variance	6.577040e+07	0.000823
Stdev	8.109895e+03	0.028697
Skewness	0.000000e+00	13.976188
Kurtosis	-1.200128e+00	288.874338

Variance shows that most of the values are tightly packed.

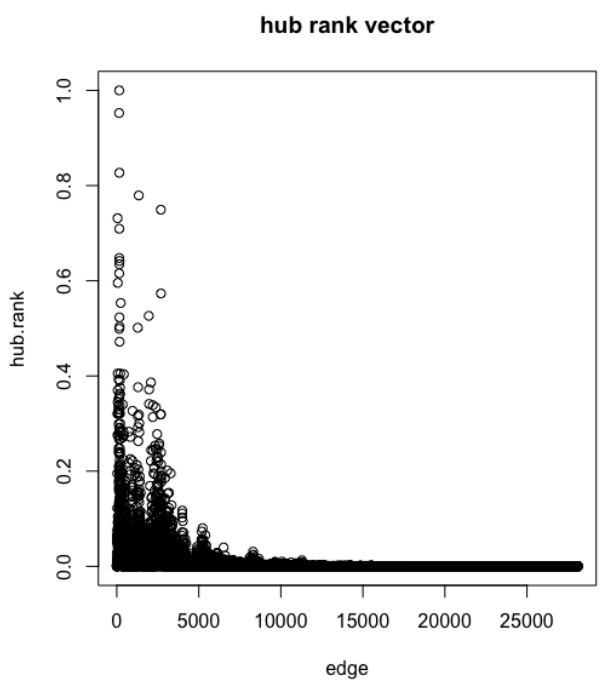

Fig. 13. Hub Rank Vector Scatter Plot

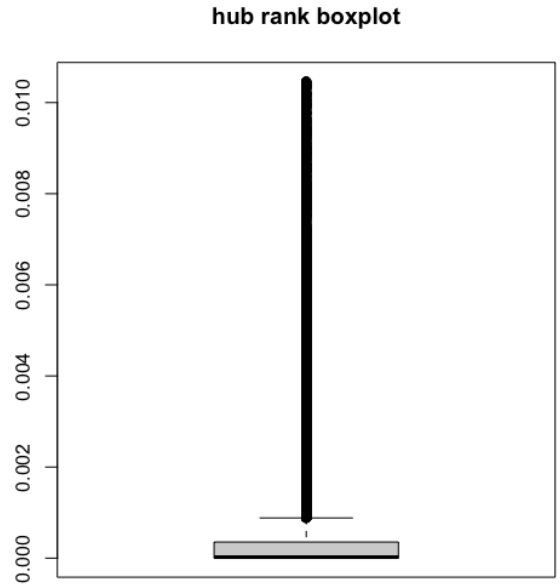

Fig. 14. Hub rank Boxplot

The box plot depicts that most of value do not have much outlinks

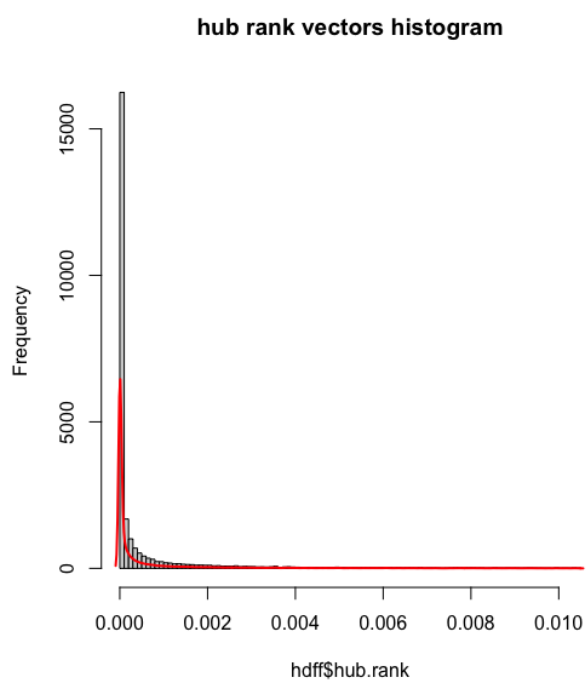

Fig. 15. Hub Rank Histogram

Hub rank boxplot also support our assertion that the data is tightly packed

We will next fit a distribution for the curve.

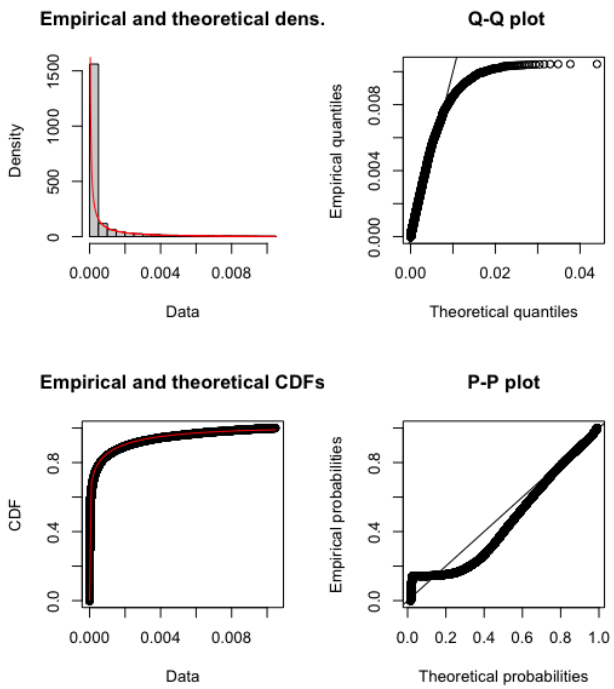

Fig. 16. Hub Rank Distribution Fit

The values shows that the vectors follow gamma with shape 0.1062302 0.0006912888 rate 155.3172988 3.0488480187

8 Conclusions

The above correlation shows us that the page HITs is the better algorithm in the case for the citation network dataset and from the EDA we can show that most of the value are only hubs or authority and very few of them are good as both authority and hubs

Acknowledgements

For this project i want to extend my tanks to our professor Dr. Kalyani Desikan for her continued guidance throughout the span of our project and also to Dr B. Jaganathan's thesis paper which motivated for this paper

References

[1] @inproceedings title=The Network Data Repository with Interactive Graph Analytics and Visualization, author=Ryan A. Rossi and Nesreen K. Ahmed, booktitle=AAAI, url=https://networkrepository.com, year=2015

[2] 'The PageRank Citation Ranking: Bringing Order to the Web" - Sergey Brin, Larry Page

[3] Jon M. Kleinberg, Ravi Kumar, Prabhakar Raghavan, Sridhar Rajagopalan, and Andrew S. Tomkins, The Web as a graph: measurements, models and methods, Proc. Fifth Ann. Int. Computing and Combinatorics Conf., SpringerVerlag Lecture Notes in Computer Science 1627, 1999, 1-17.

[4] "Modeling and Optimizing Hypertextual Search Engines" Based on the Reasearch of Larry Page and Sergey Brin, Yunfei Zhao Department of Computer Science, University of Vermont Slides from Spring 2009 Presenter: Michael Karpeles

[5] Google and the Page Rank Algorithm, slides by Székely Endre 2007. 01. 18

[6] The Intelligent Surfer: Probabilistic Combination of Link and Content Information in PageRank Matthew Richardson Pedro Domingos Department of Computer Science and Engineering University of Washington Box 352350 Seattle, WA 98195-2350, USA mattr, pedrod@cs.washington.edu

[7] Enhanced Web Page Ranking Methods for Improving Web Search by B Jaganathan